

# Algorithm: ES-MDA with Adaptive Student- $t$ Likelihood

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## Abstract

This document presents the complete algorithm for the adaptive Student- $t$  ensemble smoother described in the accompanying paper. The algorithm extends the standard ES-MDA framework with three components: (1) latent per-observation precision weights  $\lambda_i$  drawn from a Gamma distribution, implementing the scale-mixture-of-normals representation of the Student- $t$  likelihood; (2) adaptive estimation of the degrees of freedom  $\nu$  from the ensemble residuals at each iteration via kurtosis matching; and (3) inflation of both the observation perturbation by  $1/\sqrt{\lambda_i}$  and of the noise covariance matrix by  $1/\lambda_i$ , preserving the exactness of the ensemble Kalman update conditional on the latent weights. The algorithm adds approximately 50 lines to an existing ES-MDA implementation and introduces no new tuning parameters beyond those already required by the base method.

*Keywords:* ensemble smoother, Student- $t$ , pseudocode, algorithm, ES-MDA.

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## 1. Algorithm Description

The algorithm extends the standard ES-MDA loop with three insertions, marked **(S1)**–**(S3)** in Algorithm 1. The remaining steps are unchanged from the Gaussian ES-MDA formulation of Emerick and Reynolds (2013).

### **(S1) Adaptive $\nu$ Estimation**

After the forward model has been evaluated for all ensemble members (Step 2a), the ensemble-mean residuals are standardised by the current noise estimate  $\bar{\phi}_i$ . A robust sample kurtosis  $\hat{\kappa}$  is computed from residuals clipped to  $\pm 8$  standard deviations, preventing individual extreme outliers from dominating the fourth moment. If  $\hat{\kappa} > 3.1$  (indicating heavier-than-Gaussian tails), the degrees of freedom are estimated via the method-of-moments formula  $\hat{\nu} = 4 + 6/(\hat{\kappa} - 3)$ , which is exact for a Student- $t$  distribution with  $\nu > 4$ . Otherwise  $\hat{\nu}$  is set to a high value (100), reflecting approximately Gaussian behaviour. The estimate is smoothed with an exponential moving average ( $\eta = 0.7$ ) and clamped to  $[3, 100]$ . At the first iteration,  $\nu$  is initialised to a user-specified value (default 8.0) since residuals from the prior ensemble are unreliable.

### **(S2) Latent Precision Weights**

Each observation  $i$  receives a latent precision multiplier  $\lambda_i$ , drawn from a Gamma distribution. At iteration 1, the prior  $\text{Gamma}(\nu/2, \nu/2)$  is used. At subsequent iterations, the posterior conditional on the current standardised residual  $r_i$  is used:  $\lambda_i \mid r_i \sim \text{Gamma}((\nu +$

$1)/2, (\nu + r_i^2)/2$ ). This posterior draw targets down-weighting at observations that currently have large ensemble-mean residuals, rather than relying on random draws. The parametrisation  $\text{Gamma}(\nu/2, \nu/2)$  gives  $\mathbb{E}[\lambda_i] = 1$ , so typical (well-fit) observations experience standard perturbation variance.

### (S3) Perturbation and Covariance Scaling

The latent weights enter the ES-MDA machinery in two places, both required for statistical consistency with the scale-mixture representation. First, the observation perturbation is scaled by  $1/\sqrt{\lambda_i}$  (Step 2d), so that observations with  $\lambda_i \ll 1$  receive proportionally larger perturbation noise. Second, the observation noise covariance matrix  $C_d$  is inflated by  $1/\lambda_i$  (Step 2e), increasing the effective observation error variance for outliers and shrinking their contribution to the Kalman gain. Both scalings use the same weight vector  $\mathbf{w} = 1/\sqrt{\max(\boldsymbol{\lambda}, \epsilon)}$  with  $\epsilon = 10^{-6}$  for numerical stability.

Conditional on  $\boldsymbol{\lambda}$ , the likelihood is exactly Gaussian, so the Kalman algebra (covariance computation, subspace inversion, parameter update) is unchanged. The IKEA Laplace noise estimation (Step 2f) operates on raw residuals and is unaffected by the latent weights, maintaining the separation between baseline noise estimation ( $\phi$ ) and outlier accommodation ( $\boldsymbol{\lambda}$ ).

## 2. Pseudocode

## 3. Implementation Notes

### 3.1. Inversion Type Compatibility

The perturbation and covariance scaling (S3) must be applied to all inversion types supported by the base ES-MDA implementation: `efast_subspace`, `fast_subspace`, `subspace`, `svd`, `esmda`, `esmda_dask`, and `dask`. For the subspace-family methods,  $\mathbf{w}^2$  enters the per-member noise vector  $\mathbf{a}_{C_d} = (N_e - 1) \alpha_t \phi_{:,j}^2 \odot \mathbf{w}^2$ . For the eFAST and Dask methods, the condensed noise vector  $\phi_{\text{rand}}$  is multiplied element-wise by  $\mathbf{w}$ . For the SVD method, the diagonal of  $\mathbf{C}_d$  is multiplied by  $\mathbf{w}^2$ .

### 3.2. Numerical Guards

- $\boldsymbol{\lambda}$  is clipped to  $[10^{-6}, \infty)$  before computing  $\mathbf{w}$  to prevent division by zero.
- Residuals are clipped to  $\pm 8$  before kurtosis computation to robustify the  $\nu$  estimator against individual extreme outliers.
- $\nu$  is clamped to  $[3, 100]$ : the lower bound ensures the Gamma shape parameter  $\nu/2$  remains  $\geq 1.5$ , preventing degenerate draws; the upper bound provides a smooth transition to the Gaussian limit.
- The EMA smoothing constant  $\eta = 0.7$  was chosen heuristically and has not been systematically optimised. Lower values produce faster adaptation but more iteration-to-iteration variation in  $\nu$ .

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**Algorithm 1** ES-MDA with Adaptive Student- $t$  Likelihood

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**Require:** Prior ensemble  $\mathbf{M}_0 \in \mathbb{R}^{N_m \times N_e}$ , observations  $\mathbf{y} \in \mathbb{R}^{N_d}$ , inflation factors  $\alpha_1, \dots, \alpha_{N_a}$ , initial  $\nu_0 = 8$ , smoothing  $\eta = 0.7$

**Ensure:** Posterior ensemble  $\mathbf{M}_{N_a}$ , final  $\nu$ , final  $\phi$

- 1:  $\phi \leftarrow \text{HalfNormal}(\mathbf{y}, \text{error})$   $\triangleright$  per-group noise prior
- 2:  $\nu \leftarrow \nu_0$
- 3: **for**  $t = 1$  **to**  $N_a$  **do**
- 4:    $\tilde{\mathbf{M}} \leftarrow \text{inv\_transform}(\mathbf{M})$   $\triangleright$  to physical parameter space
- 5:    $\mathbf{D} \leftarrow \text{fill\_ensemble}(\tilde{\mathbf{M}})$   $\triangleright \mathbf{D} \in \mathbb{R}^{N_d \times N_e}$
- 6:   **(S1) Estimate  $\nu$  (if  $t > 1$ )**
- 7:      $\bar{\mathbf{r}} \leftarrow \mathbf{y} - \text{mean}(\mathbf{D}, \text{axis} = 1)$
- 8:      $\hat{\mathbf{r}} \leftarrow \bar{\mathbf{r}} / \max(\text{mean}(\phi, \text{axis} = 1), 10^{-8})$
- 9:      $\hat{\mathbf{r}} \leftarrow \text{clip}(\hat{\mathbf{r}}, -8, 8)$
- 10:      $\hat{\kappa} \leftarrow \mathbb{E}[\hat{\mathbf{r}}^4] / \mathbb{E}[\hat{\mathbf{r}}^2]^2$
- 11:     **if**  $\hat{\kappa} > 3.1$  **then**  $\hat{\nu} \leftarrow 4 + 6/(\hat{\kappa} - 3)$  **else**  $\hat{\nu} \leftarrow 100$
- 12:      $\nu \leftarrow \eta \nu + (1 - \eta) \text{clip}(\hat{\nu}, 3, 100)$
- 13:   **(S2) Draw latent precision weights  $\lambda$**
- 14:     **if**  $t = 1$  **then**
- 15:        $\lambda \leftarrow \text{Gamma}(\text{shape} = \nu/2, \text{rate} = \nu/2)$   $\triangleright$  prior
- 16:     **else**
- 17:        $\mathbf{r}^2 \leftarrow \hat{\mathbf{r}}^2$
- 18:        $\lambda \leftarrow \text{Gamma}(\text{shape} = (\nu + 1)/2, \text{rate} = (\nu + \mathbf{r}^2)/2)$   $\triangleright$  posterior
- 19:   **(S3) Compute perturbation scaling**
- 20:      $\mathbf{w} \leftarrow 1 / \sqrt{\max(\lambda, 10^{-6})}$
- 21:   **Perturb observations**
- 22:     **for**  $j = 1$  **to**  $N_e$  **do**
- 23:        $\tilde{\mathbf{D}}_{:,j} \leftarrow \mathbf{y} + \sqrt{\alpha_t} \phi_{:,j} \odot \mathbf{w} \odot \mathcal{N}(\mathbf{0}, \mathbf{1})$
- 24:   **Kalman update**
- 25:      $\Delta \mathbf{D} \leftarrow \mathbf{D} - \text{mean}(\mathbf{D}, \text{axis} = 1)$
- 26:      $\Delta \mathbf{M} \leftarrow \mathbf{M} - \text{mean}(\mathbf{M}, \text{axis} = 1)$
- 27:      $\mathbf{C}_{md} \leftarrow \Delta \mathbf{M} \Delta \mathbf{D}^\top / (N_e - 1)$
- 28:      $\mathbf{C}_{dd} \leftarrow \Delta \mathbf{D} \Delta \mathbf{D}^\top / (N_e - 1)$
- 29:     **for**  $j = 1$  **to**  $N_e$  **do**
- 30:        $\mathbf{C}_d \leftarrow \text{diag}(\phi_{:,j}^2 \odot \mathbf{w}^2)$   $\triangleright$  inflated by  $1/\lambda$
- 31:        $\mathbf{K} \leftarrow \mathbf{C}_{dd} + \alpha_t \mathbf{C}_d$
- 32:        $\mathbf{M}_{:,j} \leftarrow \mathbf{M}_{:,j} + \mathbf{C}_{md} \mathbf{K}^{-1} (\tilde{\mathbf{D}}_{:,j} - \mathbf{D}_{:,j})$
- 33:   **Update  $\phi$  (IKEA Laplace)**
- 34:      $\phi \leftarrow \text{covariance\_matrix}(\mathbf{D}, \text{obs\_data}, \alpha_t, \dots)$
- 35:   **Report:**  $\text{mean}(\phi)$ ,  $\nu$ ,  $\text{mean}(\tilde{\mathbf{M}}, \text{axis} = 1)$
- 36: **end for**
- 37: **return**  $\mathbf{M}$ ,  $\nu$ ,  $\phi$

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### 3.3. Computational Cost

The Student-*t* extension adds three vector operations per iteration: a Gamma draw of length  $N_d$  ( $\sim 2$  ms for  $N_d = 11,323$ ), a kurtosis computation (one pass over  $N_d$  residuals), and an EMA update (scalar). Total overhead is under 0.1 seconds per iteration, compared to 8–12 seconds for the forward model evaluations and Kalman update. The asymptotic complexity of the ES-MDA loop is unchanged.

### 3.4. Relationship to the Gaussian Limit

As  $\nu \rightarrow \infty$ , the Gamma distribution concentrates at its mean,  $\lambda_i \rightarrow 1$  for all  $i$ , and  $\mathbf{w} \rightarrow \mathbf{1}$ . The algorithm reduces exactly to the standard Gaussian ES-MDA. This property enables a single code path for both likelihoods, with the Gaussian case recovered by setting  $\mathbf{w} = \mathbf{1}$  (equivalent to `likelihood="gaussian"` in the implementation).

## References

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